

1 Upper estimate for the Heisenberg group.

Consider the Heisenberg group G and its standard matrix representation (which is faithful). Hereinafter, we denote by $\ell(x)$ length of an element $x \in G$. We use the following hypothesis (hereinafter, we forget about additive constants):

1.1. Hypothesis. Suppose that an element $x \in G$ corresponds to $\begin{pmatrix} 1 & n & k \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$; then

$$\ell(x) \geq \min_{1 \leq t \leq k} |t| + |k/t| + |m - t| + |n - k/t|. \quad (1)$$

Hypothesis ?? is based on the following Claim:

1.2. Claim. Suppose that an element $x \in G$ corresponds to $\begin{pmatrix} 1 & n & k \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$; then

$$\ell(x) \leq \min_{1 \leq t \leq k, t|k} |t| + |k/t| + |m - t| + |n - k/t|$$

(hereinafter, $t | k$ means that t is a divisor of k).

Proof. Consider $t \in \mathbb{Z}$ such that $k/t \in \mathbb{Z}$. Consider the following representation of x :

$$\begin{pmatrix} 1 & n - k/t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k/t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & m - t \\ 0 & 0 & 1 \end{pmatrix}.$$

This representation implies Claim ?? immediately. $\square$

Note that estimate (??) implies an upper estimate for a number of points in $B(r, G)$ (hereinafter, $B(r, G)$ denotes the ball of radius r in G).

Fix an arbitrary $x \in G$ and assume that it corresponds to $\begin{pmatrix} 1 & n & k \\ 0 & 1 & m \\ 0 & 0 & 1 \end{pmatrix}$. Without loss of generality we assume that $m, n, k \geq 0$. Four cases are possible.

Case 1. $n - k/t \geq 0$, $m - t \geq 0$, i.e. $k/n \leq t \leq m$. In particular,

$$k \leq mn.$$

The right side of (??) is rewritten as follows:

$$t + k/t + n - k/t + m - t = m + n.$$

Note that clearly the previous formula is the best possible lower estimate for length. Consequently in the other 3 cases we may assume that $k \geq mn$.

Let us calculate the corresponding integral:

$$\int_{0 \leq k \leq mn, 0 \leq m+n \leq r} dm dn dk = \int_{0 \leq m+n \leq r} m n d m d n = \int_{0 \leq m \leq r} \int_{0 \leq n \leq r-m} m n d m d n =$$

$$\begin{aligned}
&= \int_{0 \leq m \leq r} \frac{(r-m)^2 m}{2} dn = \int_{0 \leq m \leq r} \frac{r^2 m}{2} + \frac{m^3}{2} - rm^2 dm = \\
&= \frac{r^4}{4} + \frac{r^4}{8} - \frac{r^4}{3} = (3/8 - 1/3)r^4 = \frac{r^4}{24}.
\end{aligned} \tag{2}$$

Case 2. $n - k/t \leq 0$, $m - t \geq 0$, i.e. $t \leq k/n$, $t \leq m$.

Thus we have $k \geq mn$, $t \leq m \leq k/n$.

The right side of (??) is rewritten as follows:

$$t + k/t + k/t - n + m - t = 2k/t - n + m.$$

Clearly the minimum value of this expression is for $t = m$. It is equal to

$$2k/m - n + m.$$

Case 3. $n - k/t \geq 0$, $m - t \leq 0$, i.e., $t \geq k/n$, $t \geq m$.

Since we are interested in the case $k \geq mn$, we have $m \leq k/n$.

The right side of formula (??) is rewritten as follows:

$$t + k/t + n - k/t + t - m = 2t - m + n.$$

Clearly, in order to calculate the minimum of this expression we need to put $t = k/n$.

The minimum is

$$2k/n - m + n.$$

Consideration of this case does not give anything new.

Case 4. $n - k/t \leq 0$, $m - t \leq 0$, i.e., $m \leq t \leq k/n$. Note that we have

$$mn \leq k.$$

The right side of formula (??) is rewritten as follows:

$$t + k/t + k/t - n + t - m = 2t + 2k/t - n - m. \tag{3}$$

Case 4a. $m \leq \sqrt{k} \leq k/n$, i.e. $m \leq \sqrt{k}$, $n \leq \sqrt{k}$.

In this case the minimum value of (??) is attained for $t = \sqrt{k}$ and it is equal to

$$4\sqrt{k} - n - m.$$

Note that

$$4\sqrt{k} - n - m \leq 2k/n - m + n, \quad 4\sqrt{k} - n - m \leq 2k/m - n + m.$$

Indeed, for example, the second formula is equivalent to

$$2m + 2k/m - 4\sqrt{k} = 2(\sqrt{m} - \sqrt{k/m})^2 \geq 0.$$

The corresponding integral is

$$\int_{4\sqrt{k}-n-m \leq r, m \leq \sqrt{k}, n \leq \sqrt{k}} dmdndk = (\text{put } z = \sqrt{k}) = \int_{4z-n-m \leq r, m \leq z, n \leq z} 2zdzdmdn =$$

Using symmetry allows us to rewrite the integral as follows:

$$\begin{aligned}
&= \int_{n \leq z \leq (r+m+n)/4, m \leq n} 4zdzdmdn = \\
&\text{(note that } m \leq n \leq (r+m)/3 \text{ implies that } m \leq r/2) \\
&= \int_{n \leq (r+m)/3, m \leq n, m \leq r/2} \frac{(r+m+n)^2}{8} - (2n^2) dmdn = \\
&= \int_{m \leq n \leq (r+m)/3, 0 \leq m \leq r/2} \frac{r^2}{8} + \frac{m^2}{8} - \frac{15n^2}{8} + \frac{rm}{4} + \frac{rn}{4} + \frac{mn}{4} dmdn = \\
&\int_{0 \leq m \leq r/2} \left(\frac{r^2}{8} + \frac{m^2}{8} + \frac{mr}{4} \right) \left(\frac{r-2m}{3} \right) - \frac{5(r+m)^3}{8 \cdot 27} + \frac{5m^3}{8} + \frac{(r+m)(r+m)^2}{9 \cdot 8} - \frac{(r+m)m^2}{8} dm = \\
&= \int_{0 \leq m \leq r/2} \frac{r^3}{24} + \frac{m^2r}{24} + \frac{mr^2}{12} - \frac{r^2m}{12} - \frac{m^3}{12} - \frac{m^2r}{6} - \frac{r^3}{4 \cdot 27} - \frac{r^2m}{36} - \frac{rm^2}{36} - \frac{m^3}{4 \cdot 27} + 5\frac{m^3}{8} - \frac{rm^2}{8} - \frac{m^3}{8} dm = \\
&= \int_{0 \leq m \leq r/2} \frac{7r^3}{8 \cdot 27} - \frac{5m^2r}{18} - \frac{mr^2}{36} + \frac{11m^3}{27} dm = \frac{7r^4}{16 \cdot 27} - \frac{5r^4}{27 \cdot 16} - \frac{r^4}{9 \cdot 32} + \frac{11r^4}{4 \cdot 27 \cdot 24} = \frac{13r^4}{1728}.
\end{aligned} \tag{4}$$

Case 4b. $\sqrt{k} \leq m \leq k/n$, i.e. $n \leq \sqrt{k} \leq m$.

In this case the minimum value of (??) is attained for $t = m$ and is equal to

$$m + 2k/m - n,$$

which coincides with Case 2.

Let us compare Cases 2 and 3. Note that

$$m + 2k/m - n \leq 2k/n - m + n$$

$$m + k/m \leq k/n + n$$

$$m - n \leq k(m - n)/mn$$

$$(m - n)(mn - k)/(mn) \leq 0.$$

We see that Case 2 is better than Case 3 if $m \geq n$, which we have by assumptions of Case 4b.

Thus we need to calculate the following integral:

$$\begin{aligned}
&\int_{n \leq \sqrt{k} \leq m \leq r, mn \leq k, m+2k/m-n \leq r} dmdndk = (\text{put } z = \sqrt{k}) = \\
&= \int_{\sqrt{mn} \leq z \leq m, n \leq m \leq r, z \leq \sqrt{(r+n-m)m/2}} 2zdzdmdn = \\
&= \int_{\sqrt{mn} \leq z \leq m, n \leq m, m \leq (r+n)/3} 2zdzdmdn + \int_{\sqrt{mn} \leq z \leq \sqrt{(r+n-m)m/2}, n \leq m, n \leq 3m-r} 2zdzdmdn = (*).
\end{aligned}$$

We start from the calculation of the first integral:

$$\begin{aligned}
&= \int_{\sqrt{mn} \leq z \leq m, n \leq m, m \leq (r+n)/3} 2z dz dm dn = \int_{n \leq m \leq (r+n)/3, n \leq r/2} m^2 - mndm dn = \\
&= \int_{0 \leq n \leq r/2} \frac{(r+n)^3}{3^4} - \frac{n^3}{3} - \frac{(r+n)^2 n}{18} + \frac{n^3}{2} dn = \\
&= \int_{0 \leq n \leq r/2} \frac{r^3}{3^4} + \frac{r^2 n}{27} + \frac{rn^2}{27} + \frac{n^3}{3^4} - \frac{n^3}{3} - \frac{r^2 n}{18} - \frac{n^3}{18} - \frac{rn^2}{9} + \frac{n^3}{2} dn = \\
&\int_{0 \leq n \leq r/2} \frac{r^3}{3^4} - \frac{r^2 n}{2 \cdot 27} - \frac{2rn^2}{27} + \frac{10n^3}{3^4} dn = \frac{r^4}{2 \cdot 3^4} - \frac{r^4}{16 \cdot 27} - \frac{r^4}{4 \cdot 3^4} + \frac{5r^4}{32 \cdot 3^4} = \frac{7r^4}{2592} \quad (5)
\end{aligned}$$

Now let us calculate the second integral in (*):

$$\begin{aligned}
&\int_{\sqrt{mn} \leq z \leq \sqrt{(r+n-m)m/2}, n \leq m, n \leq 3m-r} 2z dz dm dn = \int_{n \leq m, n \leq 3m-r, m+n \leq r} \frac{rm}{2} + \frac{nm}{2} - \frac{m^2}{2} - mndm dn = \\
&= \int_{n \leq r-m, r/2 \leq m \leq r} \frac{rm}{2} - \frac{nm}{2} - \frac{m^2}{2} dm dn + \int_{n \leq 3m-r, r/3 \leq m \leq r/2} \frac{rm}{2} - \frac{nm}{2} - \frac{m^2}{2} dm dn = (**)
\end{aligned}$$

We start from calculation of the first integral in (**):

$$\begin{aligned}
&\int_{n \leq r-m, r/2 \leq m \leq r} \frac{rm}{2} - \frac{nm}{2} - \frac{m^2}{2} dm dn = \int_{r/2 \leq m \leq r} (r-m) \left(\frac{rm}{2} - \frac{m^2}{2} \right) - \frac{m(r-m)^2}{4} dm = \\
&= \int_{r/2 \leq m \leq r} \frac{r^2 m}{2} - \frac{rm^2}{2} - \frac{rm^2}{2} + \frac{m^3}{2} - \frac{r^2 m}{4} - \frac{m^3}{4} + \frac{rm^2}{2} dm = \\
&= \int_{r/2 \leq m \leq r} \frac{r^2 m}{4} - \frac{rm^2}{2} + \frac{m^3}{4} dm = \frac{r^4}{8} - \frac{r^4}{32} - \frac{r^4}{6} + \frac{r^4}{48} + \frac{r^4}{16} - \frac{r^4}{256} = \frac{5r^4}{768} \quad (6)
\end{aligned}$$

Now let us calculate the second integral in (**):

$$\begin{aligned}
&\int_{n \leq 3m-r, r/3 \leq m \leq r/2} \frac{rm}{2} - \frac{nm}{2} - \frac{m^2}{2} dm dn = \int_{r/3 \leq m \leq r/2} \left(\frac{rm}{2} - \frac{m^2}{2} \right) (3m-r) - \frac{m(3m-r)^2}{4} dm \\
&= \int_{r/3 \leq m \leq r/2} \frac{3rm^2}{2} - \frac{3m^3}{2} - \frac{r^2 m}{2} + \frac{rm^2}{2} - \frac{9m^3}{4} - \frac{mr^2}{4} + \frac{3rm^2}{2} dm = \\
&\int_{r/3 \leq m \leq r/2} \frac{7rm^2}{2} - \frac{15m^3}{4} - \frac{3mr^2}{4} dm = \frac{7r^4}{16 \cdot 3} - \frac{7r^4}{2 \cdot 3^4} - \frac{15r^4}{2^8} + \frac{5r^4}{16 \cdot 27} - \frac{3r^4}{32} + \frac{3r^4}{8 \cdot 9} = \frac{73r^4}{20736} \quad (7)
\end{aligned}$$

Summing (??), (??), (??), (??), and (??) (we multiply by 2 the last 3 integrals because of the symmetry between m and n) and taking into consideration the symmetry factor of 8, we get the following upper estimate for $|B(r, G)|$

$$|B(r, G)| \leq 8 \cdot \left(\frac{r^4}{24} + \frac{13r^4}{1728} + \frac{2 \cdot 7r^4}{2592} + \frac{2 \cdot 5r^4}{768} + \frac{2 \cdot 73r^4}{20736} \right) \approx 0.5972r^4.$$

Since the asymptotic volume is equal to

$$AV(G) := \lim_{r \rightarrow \infty} \frac{B(r, G)}{r^n}, \quad (8)$$

where n is dimension of the group (4 for the Heisenberg group) and

$$hd(H_3) = \frac{4!}{2^4} = \frac{3}{2}$$

is homogeneous dimension of the group, the upper estimate for the asymptotic volume is about 0.8958.

1.3. Remark. *We are able to get this upper estimate only if Hypothesis ?? holds. It seems that the asymptotic volume is equal to 1, and thus it seems that Hypothesis ?? does not hold.*

2 Lower estimate for the Heisenberg group.

We use the following simple claim:

2.1. Claim. *Consider an element z^k with the matrix representation $\begin{pmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.*

Choose r such that

$$r^2 \leq k \leq (r+1)^2.$$

Then

$$\ell(z^k) \leq 5\sqrt{k} \quad (9)$$

Proof. First of all assume that $r^2 \leq k \leq r^2 + r$. Then the representation

$$z^{r^2} = \begin{pmatrix} 1 & r & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & r & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{pmatrix}$$

implies

$$\begin{aligned} \ell(z^{r^2}) &\leq 4r \leq 4\sqrt{k}, \\ |k - r^2| &\leq r \leq \sqrt{k}. \end{aligned}$$

We observe that (??) holds.

Next assume that $r^2 + r < k \leq (r+1)^2$. Then, similarly,

$$\begin{aligned} \ell(z^{(r+1)^2}) &\leq 4(r+1) \leq 4\sqrt{k} + 4, \\ |k - (r+1)^2| &\leq |r+1| \leq \sqrt{k} + 1, \end{aligned}$$

which implies (??) (the additive constant is not important, that is why we omit it; we used a very rough estimate $\ell(z^t) \leq t + 1$ and we used it for comparatively small t). $\square$

Now consider an element x given by the representation $\begin{pmatrix} 1 & m & k \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix}$ and consider an element $\bar{x}_t$ (hereinafter, $\bar{\cdot}$ corresponds to round-off) given by the representation

$$\begin{pmatrix} 1 & m & \bar{t}\frac{\bar{k}}{t} \\ 0 & 1 & n \\ 0 & 0 & 1 \end{pmatrix},$$

where $1 \leq t \leq k$.

Using the estimate

$$|k - \bar{t}\bar{k}/t| \leq |t \cdot k/t - \bar{t} \cdot k/t| + |\bar{t} \cdot k/t - \bar{t} \cdot \bar{k}/t| \leq (|k/t| + |t|)/2 + 1/4$$

and applying Claim ?? to $z^{k - \bar{t}\bar{k}/t}$, we conclude that

$$|\ell(x) - \ell(\bar{x}_t)| \leq \frac{5}{\sqrt{2}} \sqrt{(|k/t| + |t| + 1/2)}.$$

Thus we get a more refined version of Claim ??:

$$\ell(x) \leq \min_{1 \leq t \leq k} |\bar{t}| + |k/t| + |m - \bar{t}| + |n - \bar{k}/t| + \frac{5}{\sqrt{2}} \sqrt{(|k/t| + |t| + 1/2)},$$

$$\ell(x) \leq \min_{1 \leq t \leq k} |t| + |k/t| + |m - t| + |n - k/t| + 2 + \frac{5}{\sqrt{2}} \sqrt{(|k/t| + |t| + 1/2)}.$$

Fix a small number ϵ . We claim that there exists a number K such that for all $k \geq K$ and all $1 \leq t \leq k$

$$\sqrt{|t| + |k/t|} \leq \epsilon(|t| + |k/t|). \quad (10)$$

Indeed, this expression is equivalent to

$$1/\epsilon^2 \leq |t| + |k/t|.$$

Since

$$\min_{1 \leq t \leq k} |t| + |k/t| = 2\sqrt{k},$$

for $K \geq 1/(4\epsilon^4)$ we have (??). Clearly (??) implies (taking into consideration the upper estimate and the fact that it is sufficient to consider only $k \geq K$ from the asymptotical point of view)

$$\begin{aligned} & \int_{\min_{1 \leq t \leq k} |t| + |k/t| + |m - t| + |n - k/t| + \frac{5}{\sqrt{2}}(|k/t| + |t| + 1/2) \leq r} dmdndk \geq \\ & \geq \int_{\min_{1 \leq t \leq k} |t| + |k/t| + |m - t| + |n - k/t| \leq r} (1 + \epsilon)^4 dmdndk \geq (1 + \epsilon)^4 0.5972r^4. \end{aligned}$$

Thus we proved the following Theorem:

2.2. Theorem. *Let H be the Heisenberg group. Then*

$$AV(H) \geq 0.8958.$$

3 Lower estimate for the product of a generalised Heisenberg group and an abelian group.

3.1. Theorem. *Let G be a direct product of the Heisenberg group H_3 and an abelian group $\mathbb{Z}^N$. Then*

$$AV(G) \geq 0.8958.$$

Proof. Note that

$$|B(r, G)| = \int_{0 \leq s \leq r} |S(s, \mathbb{Z}^N)| |B(r-s, H)| ds. \quad (11)$$

We use the following lemma:

3.2. Lemma. *Asymptotically*

$$|S(r, \mathbb{Z}^N)| = \frac{2^N}{(N-1)!} r^{N-1}.$$

Proof. Let us give here an inductive reasoning that allows to estimate $|S(r, \mathbb{Z}^N)|$. Naturally

$$|S(r, \mathbb{Z}^N)| = c_N r^{N-1} + o(r^{N-1}).$$

Our goal is to estimate c_N . Note that

$$c_N r^{N-1} = c_{N-1} r^{N-2} + 2 \sum_{1 \leq k \leq r} |c_{N-1}(r-k)^{N-2}|.$$

Passing from the sum to integral by standard methods we conclude that asymptotically

$$c_N r^{N-1} = 2 \int_{1 \leq k \leq r} c_{N-1}(r-k)^{N-2} dk = 2c_{N-1} \frac{r^{N-1}}{N-1} + o(r^{N-1}).$$

Since $c_1 = 2$,

$$c_N = \frac{2^N}{(N-1)!}.$$

□

Thus by (??) and using the lower estimate for volume of a ball in the Heisenberg group,

$$\begin{aligned} |B(r, G)| &\geq 0.5972 \frac{2^N}{(N-1)!} \int_{0 \leq s \leq r} s^{N-1} (r-s)^4 ds = 0.5972 \frac{2^N}{(N-1)!} r^{N+4} \int_{0 \leq s \leq 1} s^{N-1} (1-s)^4 ds = \\ &= 0.5972 \frac{2^N}{(N-1)!} r^{N+4} \mathcal{B}(N, 5) = \end{aligned}$$

(hereinafter, we denote by $\mathcal{B}(m, n)$ the β -function)

$$0.5972 \frac{2^N}{(N-1)!} r^{N+4} \frac{(N-1)!4!}{(N+4)!} = 0.8958 \frac{2^{N+4}}{(N+4)!} r^{N+4}.$$

Now by (??) and since

$$hd(H_3 \times \mathbb{Z}^N) = \frac{(N+4)!}{2^{N+4}},$$

$$AV(H_3 \times \mathbb{Z}^N) \geq 0.8958,$$

which proves Theorem ??.

□

3.3. Theorem. Consider $G = H_{2N+1} \times \mathbb{Z}^M$ (where H_{2N+1} is the generalized Heisenberg group). Then

$$AV(G) \geq 0.8958.$$

Proof. Indeed, by "forgetting relations" we can represent G as $H_3 \times \mathbb{Z}^{2N-2+M}$. Clearly $B(r, G) \subset B(r, H_3 \times \mathbb{Z}^{2N-2} + M)$, hence

$$0.8958 \leq AV(H_3 \times \mathbb{Z}^{2N-2+M}) \leq AV(G).$$

which allows us to get a lower estimate for asymptotic volume of H_{2N+1} . Similarly using

$$0.8958 \leq AV(H_3 \times \mathbb{Z}^{2N+M-2}) \leq AV(H_{2N+1} \times \mathbb{Z}^M)$$

we can get a lower estimate for asymptotic volume of $H_{2N+1} \times \mathbb{Z}^M$.

□

We also need the following generalization of Lemma ??:

3.4. Lemma. Consider $H_3 \times \mathbb{Z}^N$. Then

$$S_{H_3 \times \mathbb{Z}^N}(r, e) \geq 0.5972 \cdot \frac{2^N 4!}{(N+4)!} r^{N+3}.$$

Proof. In this proof we denote by w the component in H_3 of a point in $S_{H_3 \times \mathbb{Z}^N}(r, e)$, and by z_j the corresponding component in $\mathbb{Z}_j$.

Note that

$$S_{H_3 \times \mathbb{Z}^N}(r, e) = \int_{|w|+|z_1|+\dots+|z_N|=r} dz_1 \dots dz_N dw =$$

$$= \int_{|w| \leq r-t, |z_1|+\dots+|z_N|=t, 0 \leq t \leq r} dw dz_1 \dots dz_N dt = \int_{|z_1|+\dots+|z_N|=t} B_{H_3}(r-t, e) dz_1 \dots dz_N dt = (*)$$

Using the lower estimate for volume of a ball in H_3 obtained at the end of Section 2, we conclude that

$$(*) \geq 0.5972 \int_{|z_1|+\dots+|z_N|=t} \int_{0 \leq t \leq r} (r-t)^4 dz_1 \dots dz_N dt = (**).$$

By Lemma ?? we conclude that

$$(**) = 0.5972 \int_{0 \leq t \leq r} (r-t)^4 \frac{2^N}{(N-1)!} t^{N-1} = 0.5972 \frac{2^N}{(N-1)!} \mathcal{B}(N, 5) r^{N+3} =$$

$$= \frac{0.5972 \cdot 2^N (N-1)! 4!}{(N-1)! (N+4)!} r^{N+3} = 0.5972 \cdot \frac{2^N 4!}{(N+4)!} r^{N+3}.$$

□

4 k -step nilpotent groups.

Let G be a 2-step nilpotent group. Denote by m the number of basis elements of G and by n the number of basis elements of $[G, G]$.

In G any element has a representation as

$$g = \prod_{1 \leq i \leq m} g_i^{\alpha_i} \prod_{1 \leq j \leq n, 1 \leq k < j} g_{j,k}^{\beta_{j,k}},$$

where $\{g_1, \dots, g_m\}$ is a basis of G and $\{g_{j,k}\}_{1 \leq k < j \leq n}$ is a basis of $[G, G]$. Note that this representation might not be unique. However it is unique for 2-step Carnot groups (hence, for the 2-step free nilpotent group too).

We use the following

4.1. Claim. *For any $h_1, h_2 \in G$, $m, n \in \mathbb{Z}$*

$$[h_1, h_2]^{mn} = [h_1^m, h_2^n].$$

Let us calculate an upper estimate for $\ell(g_{j,k}^s)$, where $s \in \mathbb{N}$. At first, we suppose that $s = p^2$, i.e. s is an exact square. Using representation $g_{j,k}^{p^2} = [g_j^p, g_k^p]$, we observe that

$$\ell(g_{j,k}^{p^2}) \leq 4p.$$

Now suppose that

$$s = p^2 + t, \quad -p \leq t \leq p;$$

then, using the representation $g_{j,k}^s = g_{j,k}^{p^2} [g_j^t, g_k^t]$, we observe that (asymptotically)

$$\ell(g_{j,k}^s) \leq 4p + \ell(g_{j,k}^t) \leq 4p + 2|t| \leq 6p \leq 6\sqrt{s} \quad (12)$$

(for simplicity we forget about additive constants and other members of lower asymptotics).

Consider a natural embedding of G in $\mathbb{Z}^m \times \mathbb{Z}_2^n$, where $\mathbb{Z}_2^n$ is $\mathbb{Z}$ with the following nonstandard metric:

$$|(z_1, \dots, z_n)| = \sum_{i=1}^n 6\sqrt{|z_i|}.$$

Note that this embedding is surjective and is an expansion (in case of a Carnot group it is an injection).

Indeed, in order to prove that it is an expansion, it is sufficient to prove that $|g|_{\mathbb{Z}^m \times \mathbb{Z}_2^n} \geq |g|_G$ for any $g \in G$, which follows from (??).

Thus $B(r, G) \supset B(r, \mathbb{Z}^m \times \mathbb{Z}_2^n)$. We will calculate $F(m, n) := B(r, \mathbb{Z}^m \times \mathbb{Z}_2^n)$ by induction. We know that $F(1, 0) = r$ (later we take into account the multiplicative factor, 2^{m+n}). Clearly $F(m, n) = C(m, n)r^{m+2n}$. We want to find all $C(m, n)$.

Let us pass from $F(m-1, n)$ to $F(m, n)$. Note that

$$F(m, n) = \int_{0 \leq t \leq r} C(m-1, n)(r-t)^{m-1+2n} dt = \frac{C(m-1, n)}{m+2n} r^{m+2n}.$$

Let us pass from $F(m, n-1)$ to $F(m, n)$. Note that

$$F(m, n) = \int_{0 \leq 6\sqrt{t} \leq r} C(m, n-1)(r - 6\sqrt{t})^{m-2+2n} dt.$$

Put $s = 6\sqrt{t}$, then $t = s^2/6^2$, and

$$\begin{aligned} F(m, n) &= \int_{0 \leq s \leq r} 2C(m, n-1)(r-s)^{m-2+2n} s/6^2 ds = \\ &= C(m, n-1) \frac{2(m-2+2n)!}{6^2(m+2n)!} r^{m+2n} = \frac{2C(m, n-1)}{6^2(m+2n)(m+2n-1)} r^{m+2n}. \end{aligned}$$

Note that

$$C(m, n) = \frac{2^n C(m, 0)}{6^{2n}(m+2n) \dots (m+1)} = \frac{2^n C(1, 0)}{6^{2n}(m+2n)!} = \frac{2^n}{6^{2n}(m+2n)!}. \quad (13)$$

Note that we forgot the multiplicative factor of 2^{m+n} in the numerator, and we did not consider the contribution of homogeneous dimension $(m+2n)!/2^{m+2n}$ (also in the numerator). To summarize, the lower estimate for the asymptotic volume is: $1/6^{2n}$.

Now let G be a k -step nilpotent group. Denote by m dimension of G , and by $n_1, \dots, n_{k-1}$ dimensions of its commutator subgroups.

Note that Claim ?? holds for k -step Carnot groups.

Besides, we use the following:

4.2. Claim. For any $1 \leq s \leq k$, any $g_{i_1, \dots, i_s} = [\dots [g_{i_1}, g_{i_2}] \dots, g_{i_s}]$, and any $t \in \mathbb{Z}$

$$\ell(g_{i_1, \dots, i_s}^t) \leq c_s |t|^{1/s},$$

where the numbers $\{c_s\}_{1 \leq s \leq k}$ are defined by

$$c_s = 6c_{s-1} + 2, \quad c_1 = 1. \quad (14)$$

Proof. We use induction. Without loss of generality, we assume that $t > 0$.

First suppose that $|t|^{1/s} \in \mathbb{N}$. Using the representation

$$g_{i_1, \dots, i_s}^t = [g_{i_1, \dots, i_{s-1}}^{t^{1-1/s}}, g_{i_s}^{t^{1/s}}],$$

we conclude that

$$\ell(g_{i_1, \dots, i_s}^t) \leq 2c_{s-1}t^{1/s} + 2t^{1/s} = 2(c_{s-1} + 1)t^{1/s}.$$

Now let us consider the general case. Choose $p \in \mathbb{N}$ such that (asymptotically)

$$t = p^s + r, \quad -sp^{s-1}/2 \leq r \leq sp^{s-1}/2.$$

Using the representation

$$g_{i_1, \dots, i_s}^t = g_{i_1, \dots, i_s}^{p^s} [(\dots [g_{i_1}, g_{i_2}] \dots)^r, g_{i_s}],$$

we conclude that (asymptotically)

$$\begin{aligned} \ell(g_{i_1, \dots, i_s}^t) &\leq 2(c_{s-1} + 1)t^{1/s} + 2c_{s-1}r^{1/(s-1)} \leq \\ &\leq 2(c_{s-1} + 1)t^{1/s} + 2c_{s-1}((s/2)p^{s-1})^{1/(s-1)} \leq 2(c_{s-1} + c_{s-1}(s/2)^{1/(s-1)} + 1)t^{1/s} \leq \\ &\leq 2(3c_{s-1} + 1)t^{1/s}. \end{aligned}$$

□

We use the similar idea, i.e., we embed G into a product of k -groups with different metrics, i.e., to $\mathbb{Z}^m \times \mathbb{Z}_2^{n_1} \times \dots \times \mathbb{Z}_k^{n_{k-1}}$, where $\mathbb{Z}_{i+1}$ is $\mathbb{Z}$ with the following metric:

$$d_i(z_1, z_2) = c_i(|z_1 - z_2|)^{1/i}, \quad (15)$$

where c_i are defined by (??). Note that

$$\begin{aligned} c_i &= 6c_{i-1} + 2 = 6^2c_{i-2} + 6 \cdot 2 + 2 = \dots = 6^{i-1} + 2 \cdot 6^{i-2} + \dots + 2, \\ c_i &= 2(6^i - 1)/5 - 6^{i-1} = (2/5) \cdot 6^i - 6^{i-1} - 1/5. \end{aligned} \quad (16)$$

Hence

$$d_i(z_1, z_2) = (2 \cdot 6^i/5 - 6^{i-1} - 1/5)|z_1 - z_2|^{1/i}.$$

Similarly with the case $k = 2$, we introduce

$$F(m, n_1, \dots, n_{k-1}) = B(r, \mathbb{Z}^m \times \mathbb{Z}_2^{n_1} \times \dots \times \mathbb{Z}_k^{n_{k-1}}) = C(m, n_1, \dots, n_{k-1})r^{m+2n_1+\dots+kn_{k-1}}.$$

Again our goal is to calculate $C(m, n_1, \dots, n_{k-1})$.

Note that similarly with the case of $k = 2$:

$$F(m, 0, \dots, 0) = \int_{0 \leq t \leq r} C(m-1, 0, \dots, 0)(r-t)^{m-1} dt = \frac{C(m-1, 0, \dots, 0)}{m} r^m.$$

$$F(m, n_1, \dots, n_{s-1}, 0, \dots, 0) = \int_{0 \leq c_s t^{1/s} \leq r} C(m-1, n_1, \dots, n_{s-1}-1, 0, \dots, 0)(r-c_s t^{1/s})^{m+2n_1+\dots+sn_{s-1}-s} dt.$$

Put $z = c_s t^{1/s}$, then $t = z^s/c_s^s$, and

$$\begin{aligned} F(m, n_1, \dots, n_{s-1}, 0, \dots, 0) &= \int_{0 \leq z \leq r} sC(m, n_1, \dots, n_{s-1}-1, 0, \dots, 0)(r-z)^{m+2n_1+\dots+sn_{s-1}-s} z^{s-1}/c_s^s dz = \\ &= C(m, n_1, \dots, n_{s-1}-1, 0, \dots, 0) \frac{s(m+2n_1+\dots+sn_{s-1}-s)!(s-1)!}{c_s^s(m+2n_1+\dots+sn_{s-1})!} r^{m+2n_1+\dots+sn_{s-1}} = \\ &= \frac{sC(m, n_1, \dots, n_{s-1}-1, 0, \dots, 0)}{c_s^s(m+2n_1+\dots+sn_{s-1}) \dots (m+2n_1+\dots+sn_{s-1}-s+1)} r^{m+2n_1+\dots+sn_{s-1}}, \end{aligned}$$

where the numbers $\{c_s\}_{1 \leq s \leq k}$ are defined by (??).

Using similar reasoning and (??), we conclude that

$$C(m, n_1, \dots, n_{k-1}) = \frac{\prod_{1 \leq j \leq k-1} (j+1)^{n_j}}{\prod_{1 \leq j \leq k-1} ((2/5) \cdot 6^j - 6^{j-1} - 1/5)^{(j+1)n_j} (m + \sum_{1 \leq j \leq k-1} (j+1)n_j)!}.$$

Taking into consideration the symmetry factor $2^{m+\sum_{1 \leq j \leq k-1} n_j}$ and the contribution of homogeneous dimension $(m + \sum_{1 \leq j \leq k-1} (j+1)n_j)!/2^{m+\sum_{1 \leq j \leq k-1} (j+1)n_j}$, we get the following lower estimate for the asymptotic volume:

$$\frac{\prod_{1 \leq j \leq k-1} (j+1)^{n_j}}{2^{\sum_{1 \leq j \leq k-1} j n_j} \prod_{1 \leq j \leq k-1} ((2/5) \cdot 6^j - 6^{j-1} - 1/5)^{(j+1)n_j}}.$$

To summarize, we proved the following:

4.3. Theorem. *Let G be a k -step nilpotent group that can be embedded into $\mathbb{Z}^m \times \mathbb{Z}_2^{n_1} \times \dots \times \mathbb{Z}_k^{n_{k-1}}$, where $\mathbb{Z}_i$ are defined by (??). Then*

$$AV(G) \geq \frac{\prod_{1 \leq j \leq k-1} (j+1)^{n_j}}{2^{\sum_{1 \leq j \leq k-1} j n_j} \prod_{1 \leq j \leq k-1} ((2/5) \cdot 6^j - 6^{j-1} - 1/5)^{(j+1)n_j}}.$$

Remark. This technique is quite rough. For example, if we apply it to a generalized Heisenberg group H_{2n+1} , i.e., embed it in $\mathbb{Z}^{2n} \times \mathbb{Z}_2$, by (??), we get instead of 0.8958 the following lower estimate: $\frac{1}{6^{2n}}$.

5 Heisenberg group. The case when the central element is in the systol.

Let G be the Heisenberg group H_3 . Without out loss of generality, it is sufficient to consider the case when the system of generators contains 3 elements: a , b , and $c = [a, b]$, and $\ell(a) = 1$, $\ell(b) = 1$, $\ell(c) = \epsilon$ (in the general case we just consider the corresponding subgroup of the Heisenberg group). We are interested in the case $\epsilon < 1$, and we want to understand whether it allows to improve the lower estimate for the asymptotic volume or not.

Fix $k \in \mathbb{N}$ that is an exact square. Let us estimate length of c^k . Note that

$$\ell(c^k) \leq \epsilon k, \tag{17}$$

$$\ell(c^k) \leq 4\sqrt{k}. \tag{18}$$

Clearly estimate (??) is better than (??) if

$$k \geq 16/\epsilon^2. \tag{19}$$

It means that (using notations of sections 1 and 2 to represent an element of the Heisenberg group) there is a hope to get a better estimate only for k that do not satisfy (??) and arbitrary $m, n \in \mathbb{Z}$. Note that

$$B(r, e) \geq \int_{w \in \{(k, m, n) | k \geq 16/\epsilon^2\}} dw + \int_{w \in \{(k, m, n) | k \leq 16/\epsilon^2\}} dw = 0.5972r^4 + C(\epsilon),$$

where $C(\epsilon)$ is a constant that depends on ϵ and does not depend on r . Most probably we are able to get a better lower estimate for the second integral. However it will be just a smaller constant that depends on ϵ and does not depend on r . Thus, by (??),

$$AV(G) = \lim_{r \rightarrow 0} \text{hd}(G) \frac{B(r, e)}{r^4} \geq 0.8958.$$

To summarize, if the element in the systole is contained in center of the Heisenberg group, then it does not contribute to the asymptotic volume.